Dawit Chun

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Education

Taejae University Seoul, South Korea

B.S. in Data Science/AI & Natural Sciences (Physics) Expected June 2027

- GPA 4.46/4.5. Matriculation Award: admitted at age 15 as the cohort's youngest student, with the highest entrance score
- Founded in 2023, with all courses taught in English. Global rotation, one semester per city: Tokyo, San Francisco, New York, Shenzhen, Beijing (2025–2027); Vice Student Body President, 2025
- Relevant coursework: Machine Learning and Deep Learning, Artificial Intelligence I (Pace University), Vision AI for Detection and Generation of Visual Information, Computational Decision Science, Experimental Design and Statistics, Mathematics for Social and Natural Science, Intro to Digital Electronics (University of San Francisco), Electronics and Optics for Future Solutions, Physics in the Modern Society

Research Interests

My goal is to build robots that understand the world as people do, learning from what they see and from what people show them, and from their own experience as well as ours.

- Imitation learning, vision-language-action models (VLAs), World Action Models (WAMs), and foundation models for manipulation
- Reinforcement learning from human interventions
- Sim-to-real and new-embodiment transfer
- Learning in deployment: policies that generate their own corrective data and update targets
- Knowing when to trust experience: anticipating failure and asking for help

Research Experience

Research Intern **June 2026 – August 2026**

ROBOTIS, Humanoid Software Team Seoul

Video, technical story, pull requests, HG-DAgger code, Prior-ACT code

- **Off-policy RL, sim-to-real:** rebuilt a bimanual two-bottle task in Isaac Sim, calibrated to the real AI Worker, and trained a small residual actor on the frozen policy through Physical Intelligence's RL Token (TD3+BC). Real-robot success rose from 60% to 82% for Prior-ACT (50 trials) and from 17% to about 30% for TurboVLA (30 trials); on LIBERO-Spatial, online TD3 raised TurboVLA from 75% to 85% (48 episodes, reproduced)
- **Human-in-the-loop learning:** built the ROS 2 teleoperation system for HG-DAgger and HIL-SERL on the AI Worker, with auto-labeled interventions. 31 demonstrations and 13 corrected episodes taught GR00T N1.7 a long-horizon peg-and-screw task (over 30 trials, peg-in-hole 40% to 100%, driver grasp 10% to 87%, screwing 0% to 40%)
- Designed **Prior-ACT**, an ACT policy whose latent comes from a frozen DINOv3 scene prior: about 20 points higher success than vanilla ACT on all three robots (60% against 40% over the same 50 AI Worker trials), and 13–16% lower held-out action error (10 unseen episodes)
- Benchmarked four of 2026's newest VLAs and WAMs (GR00T N1.7, MolmoAct2, VLA-JEPA, FastWAM) on the 6-DOF OMY; asynchronous, age-weighted temporal ensembling raised GR00T's full-task success from 40% to 60% (10 trials per plate). Published as ROBOTIS's AI MANIPULATOR #13 video and docs
- Open source: 5 merged pull requests to ROBOTIS, including moving Cyclo Intelligence to LeRobot 0.6.1

First Author, PDPM **2025 – 2026**

Computer architecture, with Prof. Minhee Jun (Catholic University of America) Paper, code

- A RISC-V core that switches among 3-, 5-, and 7-stage pipelines at runtime, picked by a 10-weight perceptron: 94.28% accuracy at +0.107% overhead on 301 benchmarks (IEEE ICCD 2026)

Equal-Contribution First Author, iPhoneme — 2025 – 2026

Brain-computer interfaces, with Prof. Sung Jun Park (Taejae University) — arXiv, code

- Built the ConformerXL decoder for a 256-electrode speech implant: 92.14% phoneme accuracy on the T15 dataset, against 89% for Card et al. (NEJM, 2024), the prior state of the art on the same data

Research Intern — Dec 2024 – Aug 2025

Ubion Corporation, DeepCo AI education kits — Seoul

Car code, board demo video, app code (phone 1, phone 2), DeepCo Studio

- Rebuilt the DeepCo self-driving car around the Raspberry Pi 5 (PCB, servo driver, battery pack, laser distance sensor), driven by a LeNet classifier, a 2D CNN regressor, and a YOLOv8-lite detector
- Proposed the DeepCo Board, an ESP32-S3 AI education board, and built a two-floor smart-home demo on two boards for the COEX Education Fair 2025

Publications

Peer-Reviewed Papers

1. **D. Chun**, M. Jun. "PDPM: Perceptron-Driven Pipeline Morphing for Adaptive RISC-V Processors." *IEEE ICCD 2026*. **Accepted (26% acceptance).**
2. D. Chun, **D. Chun**, Y. Cha, L. Jellibi, S.J. Park. "Watch-Out: Real-Time Tracking and Assistance for Wandering Alzheimer's Patients Using Wearable Technology." *ACM CHI 2025 EA*. doi:10.1145/3706599.3720308 **1st Place (1 of 84).**
3. **D. Chun**, D. Chun, W. Hwang. "AI-Driven Smart IoT Security Window Optimization System for Disaster Response." *J. Korea Society of Computer and Information*, 30(6), 2025. doi:10.9708/jksci.2025.30.06.147
4. D. Chun, **D. Chun**, W. Hwang. "Smart Emergency Window System for Rapid Evacuation in Disasters." *J. Korea Society of Computer and Information*, 30(4), 2025. doi:10.9708/jksci.2025.30.04.093 **Excellence Paper Award.**

Preprint

1. Y. Cha*, **D. Chun***, S. Park. "iPhoneme: Brain-to-Text Communication for ALS Using ConformerXL Decoding." *arXiv:2604.16441*, 2026. *Equal contribution. doi:10.48550/arXiv.2604.16441

In Preparation

1. **D. Chun**. "IGEN: Intergenerational Affective Meta-Reinforcement Learning." MU, a robot mind for CausalWorld's TriFinger robot with a body it can lose. An earlier emotion, learned only by forecasting its own body, causally set how deeply it imagines, how cautiously it acts, and whether it keeps its grip. The next version chooses with a gut feeling learned from its own attempts; with predictions declared first, it scored 0.882 on CausalWorld pushing and 0.644 on picking on worlds never used before (designed grip), above the best published agent per protocol (0.804, 0.626), and 0.847 on Meta-World's ML45 test tasks (re-trained MAML-TRPO: 0.427).

Patents

1. **D. Chun**, D. Chun, W. Hwang. "AI-Based Integrated Smart Security Window and Automatic Control System." Korean Patent Application 10-2026-0029503, filed Feb 13, 2026.

Books (Co-author)

1. **D. Chun** et al. *Youth Conversations on Sustainable Development*. Office for Government Policy Coordination, 2025. Pub. No. 11-1092000-100007-01.
2. W. Hwang, **D. Chun**, D. Chun. *ChatGPT: Practical AI for the Classroom*. ISBN 9788970930084, 2024.

Teaching Experience

Guest Instructor — July 2026

Geumcheon Science Cube, Seoul

Taught 20 elementary students to train a self-driving car's AI (LeNet-5), from data collection to driving

Teaching Assistant, Taejae University — Fall 2025, Spring 2026

- **NS200: Physics in the Modern Society** (San Francisco): ran a weekly TA session, graded all problem sets, wrote extra thermodynamics worksheets, and helped students design mechanics experiments

- **XD221: Principles of Quantitative Thinking in the Sciences** (New York): ran weekly calculus and linear algebra sessions, taught supplementary lectures, and advised students' deep learning projects

STEM Instructor **July 2022 & 2023**

KOICA / Korea Productivity Center, Phnom Penh, Cambodia

Trained 30 high-school science teachers each summer in Arduino, 3D modeling, and C; the course materials were adopted in four further provinces

Fellowships and Training

Oxford-Stanford-Taejae Fellowship **Mar & Aug 2025**

Kellogg College, Oxford & Taejae University, Seoul Full funding ($4,000)

One of 8 Taejae representatives in Kellogg College's Sustainability Changemakers UK 2025 program, on AI governance and climate policy; won 1st place in the AI-Powered Sustainability Policy Debate in Seoul, donated the $2,000 prize to WWF Korea, and presented our recommendations to Korea's Deputy Prime Minister

DYROS Robotics Bootcamp **August 2025**

Seoul National University, Profs. Hyeongki Lee and Jaeheung Park

Selected for an intensive two-session program: control algorithms, motion planning, ROS, inverse kinematics, and actuator control

KAIST AI and Data Science Masterclass **Mar 2023 – Jan 2024**

KAIST Center for Gifted Education Grade: A+

Selected for the DICIA-KAIST AI Master Class on computer vision and algorithmic thinking

Honors & Awards

Talent Award of Korea, Ministry of Education 2025

Minister of Education Award, one of 60 recipients in the University/Young Adult category. Donated the entire 2 million won prize to the Geumcheon Future Scholarship Foundation.

1st Place (1 of 84), ACM CHI 2025 Student Design Competition, Yokohama 2025

1st Place, Oxford-Stanford-Taejae AI Sustainability Policy Debate 2025

Silver Medal, International Robot Olympiad, AI Autonomous Driving, Busan 2025

Gold Medal, International Robot Olympiad, AI Autonomous Driving, Athens 2024

Excellence Paper Award, KSCI Winter Conference 2024

Mayor's Commendation, Gangnam-gu, for volunteer work 2023

Before university (2021–2023): Gold Prize, Aramco Coding World Cup; Gold and Bronze Prizes, KSME Student Design Competition; 1st Place, Geumcheon Hackathon; KAIST President's Award. Five IRO medals in total, 2023–2025.

Skills

Programming:	Python, C++, C, VHDL, Verilog, SystemVerilog, Arduino, ESP32, Raspberry Pi
Tools:	PyTorch, TensorFlow, OpenCV, ROS 2, MuJoCo, Isaac Sim, Isaac Lab, LeRobot, W&B, SolidWorks, 3D printing, Git, LaTeX
Domains:	VLAs and WAMs, imitation and reinforcement learning, human-in-the-loop learning, sim-to-real, CPU architecture, IoT

Certifications

GPA 4.46/4.5	GRE 326 (V162, Q164)	TOEFL iBT 118/120
HSK Level 4	OPIc Advanced Low	Generative AI Expert Level 2

Volunteer Work

U9 Change (2021 to now): connecting Korean sponsors with the children they support in Kenya and South Sudan; Excellence Award, 2023. Hosanna Church meal delivery for elderly residents living alone (2021–2023); Gangnam-gu amphibian conservation (2023); Geumcheon science classrooms.